Reinforcement Learning and Test-Time Verification for Small-Model Mathematical Reasoning

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Abstract

We tackle the CSE 151B/251B math reasoning competition, where a fixed 4B-parameter model must answer 943 held-out problems scored by exact match. Two facts shape the whole project: the model (`Qwen3-4B-Thinking-2507`) cannot be swapped, and no external models may be called at inference time. We improve accuracy along two independent axes. First, we fine-tune the model with Group Relative Policy Optimization (GRPO) on 227 hard problems curated from the public set, using a token-level loss and a sign-advantage reward that keeps a gradient signal even when a 4B model gets every rollout in a group wrong. Second, we build an inference pipeline around 12-way majority voting and a selective self-verification pass that we apply only to free-form questions. On our 112-question validation split, GRPO adds $+1.8$ points, prompt fixes add another $+0.9$, and selective verification adds $+5.3$ on free-form ($+2.8$ overall), lifting overall accuracy from 61.6% (base model) to roughly 68%. Final private leaderboard score: 0.637, rank: 16/105. Code: Github; model weights: `Kevinhhhhhh/qwen3-4b-grpo-v6`.

1 Introduction

Problem Definition. The task is automatic mathematical problem solving. Given a problem stated in natural language and LaTeX, the model must return a final answer that an automated judge accepts under exact match. Problems come in two formats. Multiple-choice questions (MCQ) expect a single letter. Free-form questions expect one or more numeric or symbolic values placed in a `\boxed{}` field, with the number of expected values signalled by `[ANS]` placeholders in the prompt. The competition releases 1,566 public problems with gold answers and 943 private problems without them; we submit predictions for the private set and are scored on a held-out portion.

Problem Significance. Reliable mathematical reasoning is one of the clearest tests of whether a language model reasons or pattern-matches. Models that solve competition math also help students check work, assist scientists with derivations, and verify steps in formal proofs. The interesting constraint here is scale: we are restricted to a 4B-parameter model, roughly an order of magnitude smaller than the frontier systems that report strong math results. Squeezing competition-level accuracy out of a small model matters because small models are what run on a single GPU, in a classroom, or on a phone.

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Technical Challenge. The project is hard for five reasons. **(1) Data:** the judge is strict. A correct value formatted as two separate `\boxed{}` blocks instead of one comma-separated block counts as wrong, so format errors masquerade as reasoning errors. **(2) Model:** a 4B thinking model produces long chains of thought that routinely run to 20–40K tokens, which stresses both context length and memory. **(3) Implementation:** reinforcement learning on a model this size with 16K-token completions does not fit on a 40GB GPU under default trainer settings. **(4) Engineering:** a single inference pass over 943 problems at 12 samples each takes about 17 hours on four A100s, so every ablation is expensive. **(5) Evaluation:** the private set has no labels, so we must trust a small validation split and design choices that generalize.

Contributions. Prior small-model competition entries lean on majority voting alone and accept the base model as-is. We show that a 4B thinking model leaves accuracy on the table in two recoverable places: the policy itself, and the gap between a correct chain of thought and a correctly formatted answer. Our main contributions are:

- **A GRPO recipe that actually moves a 4B policy.** We diagnose why the standard recipe stalls (sequence-level length bias, masked truncations, group-mean gradient starvation, and over-hard training data) and fix each one, turning a flat 0.0-point run into a $+1.8$-point gain (Sections 3 and 4.4).
- **Selective self-verification.** A second model pass that re-checks only free-form answers adds $+5.3$ points on free-form while leaving MCQ untouched; applying it to MCQ instead *costs* 2.4 points (Section 4.4).
- **Judge-aware prompting.** Five format rules tied to concrete judge behaviors recover $+0.9$ points with no extra compute (Section 3).
- **A reproducible, externally-modelless pipeline** that stays within the competition rules and runs end-to-end on four A100s, released with weights and code.

2 Related Work

Reinforcement learning for reasoning. GRPO, introduced with DeepSeekMath [5], replaces the PPO value network [4] with a group-relative baseline: it samples a group of completions per prompt and centers each reward by the group mean. DeepSeek-R1 [1] scaled this idea to elicit long chains of thought directly from a reward signal. DAPO [9] reported that the sequence-level GRPO loss biases the model toward short completions and proposed a token-level normalization, which we adopt. Our setting differs from these works in scale. They tune 7B–671B models on tens of thousands of problems; we tune a 4B model on 227, which makes gradient starvation on uniform-reward groups the dominant failure mode rather than an edge case.

Test-time scaling for math. Self-consistency [6] samples many chains of thought and votes on the final answer, building on chain-of-thought prompting [7]. Majority voting is our backbone. Process reward models rerank candidates with a separate scorer, but the competition bans external models at inference, so reranking is off the table. We instead use the same model to verify its own free-form answers, which respects the rules and still corrects late-stage arithmetic slips.

3 Methods

Problem Setting. Let x be a problem and y its gold answer. A policy π_θ generates a response $o = (\text{reasoning}, \hat{y})$, from which the judge extracts $\hat{y}$ and returns a binary correctness $r(o, y) \in \{0, 1\}$ under exact match after numeric normalization. The dataset is $\{(x_i, y_i)\}$; the model ingests the problem text (plus a short format instruction) and emits a chain of thought followed by a boxed answer. We do not have a differentiable objective on r, so we optimize the policy with GRPO, which maximizes expected reward through a clipped policy-gradient surrogate. For a group of G sampled completions $\{o_1, \ldots, o_G\}$ for one prompt, the token-level objective we use (DAPO form) is

$$\mathcal{J}(\theta) = \mathbb{E}\left[\frac{1}{\sum_g |o_g|}\sum_{g=1}^{G}\sum_{t=1}^{|o_g|}\min\Big(\rho_{g,t}\,A_g,\ \text{clip}(\rho_{g,t}, 1-\epsilon, 1+\epsilon)\,A_g\Big)\right], \tag{1}$$

Algorithm 1 v7 inference pipeline (per problem)

Input: problem x, type $\in$ {MCQ, free-form}
Build prompt with judge-aware rules and `[ANS]`-count hint
Sample N=12 responses from π_θ at T=0.7, `max_tokens`=65,536
if type is free-form **then**
    Re-prompt each response for self-verification at T=0.3; replace with verified response
end if
Normalize each boxed answer with the judge's `norm_math_str`
return majority vote over the N normalized answers

where $\rho_{g,t} = \pi_\theta(o_{g,t} \mid x, o_{g,<t}) / \pi_{\theta_{\text{old}}}(o_{g,t} \mid x, o_{g,<t})$ is the per-token importance ratio and A_g is the advantage of completion g. Normalizing by total token count (rather than per sequence) removes the length bias that lets short wrong answers dominate the gradient.

Idea Summary. We attack the 61.6% base accuracy from two sides. Training-time: GRPO nudges the policy toward producing correct boxed answers on the competition distribution. Inference-time: voting and verification clean up the residual errors a tuned policy still makes. The two are complementary because they fix different failures. GRPO improves the reasoning the model commits to; verification catches a correct line of reasoning that ends in a slipped final number. The main risk is that GRPO on a tiny curated set overfits or fails to move the policy at all; our fallback, which the data forced us to lean on, is to keep the inference improvements even when a training run yields nothing (Section 4.4).

Method Description. **Judge-aware prompting.** The default thinking-model prompt produces answers the judge rejects even when the math is right, because the judge takes only the last contiguous group of `\boxed{}` entries and splits on commas. We add five rules to the system prompt: put all answers in one box; comma-separate sub-answers; wrap value-lists in parentheses; concatenate "select all" letters without separators; and report six significant figures. We also inject an `[ANS]`-count hint into each free-form prompt so the model knows how many values to box.

GRPO fine-tuning. We adapt the model with LoRA [2] (r=16, α=32, all seven projection matrices, dropout 0.05). The reward is $+1$ for a correct boxed answer, -1 for wrong or missing, and -0.5 for a completion truncated at the length limit, so truncation is discouraged without being treated as outright failure. We replace the group-mean advantage with a *sign advantage*: $A_g = r_g$ directly, with no centering. On a 4B model, easy and hard problems often produce all-correct or all-wrong groups; group-mean centering zeroes their advantage and wastes the step. Using the raw reward keeps every sample informative. We curate training data by running 12-way majority vote with the base model on the public set and keeping only the 227 problems it gets wrong, which guarantees mixed-reward groups and a real gradient.

Inference pipeline. Algorithm 1 shows the v7 pipeline. We shard the 943 problems across four GPUs, sample N=12 completions per problem at T=0.7, run a verification pass on free-form problems only, and take a normalized majority vote.

Data splits and normalization. We hold out `val_10pct.jsonl`, a 112-problem split (44 MCQ, 68 free-form) sampled from the public set, for every ablation; the rest of the public set is available for training-data curation. We do not normalize inputs; the judge's `norm_math_str` handles answer-side normalization (for example collapsing `0.5` and `1/2`) before votes are counted, so numerically equal answers do not split the ballot. The model runs in 4-bit (BitsAndBytes) to fit the 4B weights and a 65K-token KV cache on a 40GB card.

4 Experiments

4.1 Baselines

We compare against simpler configurations rather than separate architectures, since the model is fixed:

Table 1: Final inference hyperparameters. Verification and temperature choices are justified by the ablations in Section 4.5.

Parameter	Value	Rationale
Samples N	12	Best accuracy/compute tradeoff
Temperature T	0.7	Single-temp beats multi-temp
`max_tokens`	65,536	Long CoT needs full context
Verification	free-form only	$+5.3$pp FF, -2.4pp if applied to MCQ
Verify temperature	0.3	Conservative re-check
Quantization	4-bit (BitsAndBytes)	Fit 4B + KV cache on 40GB

- **Base model:** `Qwen3-4B-Thinking-2507` [8] with the default prompt and single-sample greedy-style decoding.
- **Base + voting:** the base model with $N{=}12$ majority voting and no fine-tuning or verification.
- **v5 (failed GRPO):** our first serious GRPO run, used as a negative baseline to isolate which training fixes mattered.

4.2 Evaluation

The competition scores exact match via the official judger on the private set. Locally we report the same judger's accuracy on the 112-problem validation split, broken out by MCQ and free-form, since the two formats respond differently to every intervention. All validation numbers below use $N{=}12$ at $T{=}0.7$ unless noted.

4.3 Implementation details

Platform. All training and inference ran on NVIDIA A100 40GB GPUs (four for parallel inference). One full private inference pass takes about 17 hours; the GRPO run took roughly 36 hours of wall time for about one epoch over the 227 problems.

Inference hyperparameters. Table 1 lists the final settings. We sample $N{=}12$ because $N{=}8$ and $N{=}12$ tied at 65.2% on validation while $N{=}16+$ would double wall time for marginal gain. We keep $T{=}0.7$ single-temperature after multi-temperature sampling underperformed (Section 4.5). `max_tokens` is the full 65,536 because chains of thought often exceed 20K tokens and earlier truncation produced wrong answers from correct reasoning.

Training hyperparameters. GRPO used LoRA ($r{=}16$, $\alpha{=}32$), learning rate $1\text{e}{-6}$, group size $G{=}8$, `max_completion_len`$=16{,}384$, sampling temperature $T{=}1.0$, token-level (DAPO) loss, no KL penalty ($\beta{=}0$), and `mask_truncated_completions`=False. We tuned the learning rate (Section 4.5 discusses $5\text{e}{-7}$ vs. $1\text{e}{-6}$) using the clip ratio as the diagnostic: a learning rate that leaves `clip_ratio` below $1\text{e}{-3}$ means the policy is not moving. Fitting this configuration on 40GB required three memory fixes: keeping logits in bf16, freeing the vLLM [3] KV cache before the log-probability forward pass, and gradient-checkpointing the per-completion forward so logits are recomputed one at a time.

4.4 Results

GRPO fine-tuning. Table 2 traces accuracy across versions. The base model scores 61.6% overall. Our first GRPO attempt (v5) landed at exactly the base accuracy, a total non-result. The fixed recipe (v6) reaches 63.4%, a $+1.8$-point gain that comes almost entirely from MCQ ($75.0 \rightarrow 77.3$) and free-form ($52.9 \rightarrow 54.4$). Adding the judge-aware prompts on top brings overall validation accuracy to 64.3%.

Selective verification. Table 3 reports the verification pass on the v6 model at $N{=}12$. Verifying every answer raises free-form by $+5.3$ points but drops MCQ by 2.4, because the model is already confident on MCQ and a second pass mostly introduces noise that flips correct letters. Applying verification to free-form only keeps the free-form gain and the original MCQ score, for an estimated 68.0% overall. This is the single largest inference-time lever in the project.

Table 2: GRPO model improvement on the 112-problem validation split (N=8). v5 reproduces the base score exactly; the fixed v6 recipe gains +1.8 points, and judge-aware prompts add +0.9 more.

Model	N	MCQ	Free-form	Overall
Base (`Qwen3-4B-Thinking-2507`)	8	75.0%	52.9%	61.6%
v5 (failed run)	8	75.0%	52.9%	61.6%
v6 (fixed recipe)	8	77.3%	54.4%	63.4%
v6 + judge-aware prompts	8	77.3%	55.9%	64.3%

Table 3: Verification ablation (v6, N=12, T=0.7). Verifying free-form only captures the free-form gain without the MCQ regression. (*partial-shard estimate; direction confirmed.)

Condition	MCQ	Free-form	Overall
No verification	79.5%	55.9%	65.2%
Verify all	77.1%*	61.2%*	67.9%*
Verify free-form only	**79.5%**	**61.2%**	**68.0% (est.)**

Private leaderboard. On the held-out private set this pipeline scored 63.7%, ranking 16 of 105 teams. The roughly four-point drop from our 68% validation estimate is expected: the 112-problem validation split is small and drawn from the public distribution, so it slightly overstates accuracy on the larger, harder, unseen private set.

4.5 Ablations

Single vs. multi-temperature. We tested generating the N=12 samples across four temperatures (0.4/0.7/0.9/1.1) to raise diversity. It hurt: free-form fell 3.0 points (55.9 → 52.9) while MCQ held. The cause is budget. Fitting four temperature groups forced `max_tokens` down to 32K, which truncated more chains of thought and turned correct reasoning into wrong boxed answers. Diversity did not pay for the lost context.

Which GRPO fixes mattered. The v5→v6 gap isolates four changes, each diagnosable from training curves. Raising the learning rate from 5e−7 to 1e−6 moved `clip_ratio` from a flat 2e−5 into the 5e−5–2e−4 range, the first sign the policy was learning at all. Switching the loss from sequence-level to token-level removed the length bias that had been rewarding terse wrong answers. Setting `mask_truncated_completions` to False recovered the 54–69% of completions per step that v5 had been silencing. Replacing the over-hard DAPO-Math-17k data with 227 curated public problems gave every group mixed rewards instead of uniform −1. Removing any one of these reproduces the v5 stall.

5 Discussion

We improved a fixed 4B math model from 61.6% to roughly 68% validation accuracy by combining a corrected GRPO recipe with a voting-and-verification inference pipeline, all without external models.

Achievements. We contributed on three fronts. On algorithms, we identified and fixed four concrete reasons GRPO stalls on a small model and turned a dead run into a +1.8-point gain. On inference, we showed that asymmetric self-verification, applied to free-form but not MCQ, is worth +5.3 points on the harder format. On data, we curated a 227-problem training set whose only selection rule, "the base model gets this wrong under voting," is exactly what GRPO needs to see a gradient. The released model and pipeline reproduce these numbers end-to-end.

Bottlenecks and Mitigation. The hard bottleneck was memory. A 4B model with 16K-token completions and an 8-way group does not fit on 40GB under the stock trainer, which simply OOMs. We wrote a memory-efficient trainer that keeps logits in bf16, frees the generation KV cache before the log-probability pass, and gradient-checkpoints the per-completion forward; together these brought peak usage just under 40GB. The second bottleneck was iteration cost: at 17 hours per private pass,

we could not afford many full runs, so we did all design selection on the 112-problem validation split and ran the private set only for the final submission. Compared to the milestone, the largest change was abandoning external-data GRPO (DAPO-Math-17k) once we traced its flat reward to data that was simply too hard for a 4B model.

Lessons Learned. The recurring lesson was that the failure was usually upstream of where it showed up. A flat validation score traced back to a training curve, and the training curve traced back to a single mis-set flag or a dataset calibrated for a 32B model. Reading the clip ratio and reward mean told us more than re-running ever did. On problem selection, we learned to respect the judge: several apparent reasoning errors were format mismatches, and writing five prompt rules recovered points we had been losing to punctuation. On literature, the DAPO and DeepSeekMath papers gave us the token-level loss and the group-baseline framing, but the small-model regime forced the sign-advantage modification they do not need. Given more time, we would test higher LoRA rank, a second GRPO epoch from the v6 checkpoint, and a learned stopping rule for chains of thought to cut the truncation rate.

Team Responsibilities. Kevin Huang built the GRPO trainer and the verification system in inference. Evelyn Zhang conducted rigorous hyperparameter tuning, performed error analysis on validation sets. Ziyao Zhou built the initial vLLM baseline, and helped with project evaluation, documentation, and managed project milestones.

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